

Proper-time theory applied to IBM-quantum coherence: a microwave-coupling angle/shape research arc

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Proper-time theory on IBM quantum coherence — microwave-coupling angle/shape focus

Date: 2026-05-31. **From:** Trey Morris (with Claude Opus 4.7). **Re:** Three-phase research arc proposal; audience default *internal-campaign planning* (see Asks). Tracks against [issue #103](#). Builds on the dual Maxwell verification (Eq. (4)), see [Two_Mathematically_Equivalent_Versions_of_Maxwells_Equations](#) and the precision-atomic campaigns [#50](#), [#78](#), [#82](#).

Page 1 — the apparatus-side hook

What is known apparatus-side. Gill’s proper-time formulation produces a dual Maxwell wave equation, verified as Eq. (4) of [Two_Mathematically_Equivalent_Versions_of_Maxwells_Equations.md](#), that differs from the textbook form by a single first-derivative term. The verification doc records the discriminating statement at [line 232](#): Eq. (4) reproduces the standard wave equation under $c \rightarrow b$, $t \rightarrow \tau$, $\mathbf{w} \rightarrow \mathbf{u}$, **plus** an extra dissipative term whose coefficient is $-(\mathbf{u} \cdot \mathbf{a})/b^4$ in front of $\partial_\tau \mathbf{E}$ (or $\partial_\tau \mathbf{B}$). The same structure resurfaces as the third term of the modified Liénard–Wiechert fields at [line 332](#): “the third term is new and tied to the dissipative coefficient $(\mathbf{u} \cdot \mathbf{a})/b^4$.” The term vanishes identically when $\mathbf{a} = \mathbf{0}$ (source unaccelerated) and inherits an angular dependence through $\mathbf{u} \cdot \mathbf{a} = |\mathbf{u}||\mathbf{a}|\cos\theta$ and a temporal/shape dependence through $\mathbf{a}(\tau)$. This is the *only* structural deviation Gill’s classical-field theory makes from textbook Maxwell at this order; every other dual modification is a $c \rightarrow b$ substitution or a kinematic re-parametrisation.

Why it might bear on transmon-qubit coherence. A transmon-qubit gate operation drives a Cooper-pair condensate with a shaped microwave pulse $A(\tau)$ along a specific drive axis $\hat{\mathbf{d}}$, so the source is *accelerated by construction* during gate operations and the framework’s extra term *can* contribute. The substantive open question — the core of Phase A — is which condensate kinematic quantity plays the role of $\mathbf{u}/\mathbf{a}$ in a transmon: candidate identifications include (i) the supercurrent velocity $\mathbf{u}_s \propto \nabla\varphi$ from the phase gradient, (ii) the AC-Josephson velocity at the drive frequency, or (iii) the time-derivative of the Cooper-pair dipole moment. The three give different effect-size estimates: at Cooper-pair drift speed $\beta^4 \lesssim 10^{-30}$ (unmeasurable), at Fermi speed $\beta^4 \sim 10^{-12}$ (below current T_2 floor by ~ 6 orders), at AC-Josephson speed it depends on drive amplitude. **Falsifiable hypothesis (Phase B):** shaping the microwave coupling to minimise $(\mathbf{u} \cdot \mathbf{a})/b^4$ should produce an angle/shape signature *distinct from* the standard Abraham–Lorentz–Dirac radiation-reaction

term ($\propto \dot{\mathbf{a}}$); the two are in-principle separable by varying θ at fixed $\dot{\mathbf{a}}$, and any measured angle dependence not accounted for by ALD is a falsifier in the direction of (or against) the framework.

Page 2 — three-phase research arc

Phase A – apparatus mapping. Derive the dissipative-term contribution to the transmon-qubit effective drive Hamiltonian (or equivalent Lindblad operator) starting from Eq. (4), *after* committing to one of the three candidate $\mathbf{u}/\mathbf{a}$ identifications above (or a fourth proposed by Tepper). Acceptance: a closed-form (or numerical-but-tractable) expression for the noise contribution as a function of θ and $A(\tau)$ with explicit $b \rightarrow c$ and $\beta \rightarrow 0$ limits, validated numerically against a no-framework baseline via `qiskit-dynamics` (the `Solver` integrates the Lindblad equation under the envelope $A(\tau)$ entered as a time-dependent `Signal`), and a numeric $\Delta T_2/T_2$ at IBM Heron’s nominal drive parameters (anharmonicity ~ 300 MHz, $t_g \sim 30$ ns). **Null-result-most-likely closure:** if $\Delta T_2/T_2 < 10^{-3}$ at the highest computable order, or the contribution factorises into the standard ALD account with no residual angle dependence separable from $\dot{\mathbf{a}}$, the program closes honestly at Phase A.

Phase B – concrete predictions. Conditional on Phase A producing a nonzero residual, specify drive angles and pulse-shape envelopes where the dissipative-term contribution is maximised vs. minimised at fixed drive amplitude. Acceptance: at least one explicit prediction of the form “*varying θ from 0 to $\pi/2$ at fixed amplitude A_0 and gate time t_g changes the framework-predicted T_2 contribution by $\Delta T_2 = \dots$, with a sign,*” with the predicted effect size compared against state-of-the-art transmon T_2 ($\sim 100\text{--}300$ μs on IBM Eagle/Heron) and against the current IBM-pulse-control angular resolution.

Phase C – IBM Quantum test. Specify the IBM Quantum Network resources: a single high-coherence transmon on Eagle- or Heron-class hardware; DRAG/cosine pulse-shape control; angular control via the standard IQ-modulation plane. Pre-flight in simulation via `qiskit-aer NoiseModel.from_backend()` augmented with a custom Lindblad channel for the framework’s term (equivalently, `qiskit_dynamics.backend.DynamicsBackend` carrying the same operator); on-hardware execution via `qiskit-ibm-runtime Sampler` under `qiskit-experiments T2Hahn/T2Ramsey` protocols, with custom drive shapes via `qiskit.pulse` *if* pulse-level access is provisioned (otherwise gate-level DRAG-coefficient sweeps). The discriminating observable is the *angle* dependence at fixed DD sequence, not raw T_2 — CPMG/Uhrig suppress low-frequency dephasing and would mask any framework contribution with overlapping spectral support, so the framework’s signature is measured as a residual after subtracting matched-DD baselines. Acceptance verdicts: [OK] framework predicts the measured angle/shape residual within the campaign’s precision, [warn] framework matches at precision floor only (consistent but not discriminating), [X] framework refuted by absence of the predicted signature at the predicted sensitivity.

Page 3 — methods, asks, honest scope

Methods (incremental vs. standard). The table below names what is already standard practice for transmon-coherence extension, alongside what the framework would *additionally* prescribe *if* Phase A succeeds.

Standard transmon-coherence technique	What the framework adds (if Phase A succeeds)
DRAG-optimised pulse shaping (leakage suppression)	Angle/shape-modulated drive minimising $(\mathbf{u} \cdot \mathbf{a})/b^4$ at fixed leakage.
Dynamical decoupling (CPMG / Uhrig / DCG)	Proper-time-spaced CPMG; DCG variants minimising $\int (\mathbf{u} \cdot \mathbf{a})/b^4 d\tau$ at fixed fidelity.
Numerical baseline (<code>qiskit-aer</code> + <code>NoiseModel.from_backend()</code>)	Same noise model with a custom Lindblad channel from the Phase-A operator, sim-only $\Delta T_2(\theta)$.

Modeling + encoding pre-flight (Qiskit). End-to-end laptop simulation analogue before any IBM credit is burned: Phase A in `qiskit-dynamics` (Lindblad `Solver` driven by $A(\tau)$ as a `Signal`); Phase B as a swept- θ batch on `qiskit-aer` producing the predicted $T_2(\theta)$ ratio that hardware will reproduce or refute; Phase C via the noise-augmented `Aer/DynamicsBackend` under the same `qiskit-experiments` protocol as the hardware run. Encoding-sequence candidates worth flagging in parallel: proper-time-spaced CPMG (inter-pulse interval set in τ , sensitive to b/c at the precision floor), and DCG variants whose envelope minimises the framework’s angle/shape integral at matched gate fidelity.

Asks (audience-dependent). Default audience at draft-time is *internal-campaign planning*; the four candidate audiences listed in [issue #103](#) (Audience options section) are equally compatible with the body above. Asks under the default audience: (a) Trey’s sign-off on the apparatus-mapping move on Page 1; (b) a superconducting-qubit theorist for Phase-A vetting (identification of Cooper-pair $\mathbf{u}/\mathbf{a}$ in the transmon charge basis); (c) optionally Tepper Gill or one of his students for the framework-internal Wolfram-MCP derivation of the Phase-A expression; (d) IBM Quantum Network compute time scoped only after Phase B produces a concrete prediction worth burning calibration on. Cross-linked to the framework-specs thread [#75](#) and the open proper-time one-loop frontier [#55](#).

Honest scope (load-bearing). This white paper proposes a *falsifiable research arc*. It does not claim a *derived prediction*. The framework’s existing validation (Bethe-Salpeter [#50](#); Li ²⁺ Z-extension [#78](#); hydrogenic g-factor Z-scan [#82](#) with Outcome C — cutoff has no Z-dependence and the framework inherits QED’s bound-state structure) is at a precision at which the framework reproduces textbook QED *by construction*. Extending from bound electrons in atoms to a superconducting transmon-qubit condensed-matter device is a substantive conceptual leap, and the most likely outcome at Phase A is *null* under the explicit threshold above. In that case the program closes honestly at Phase A and the campaign records the closure as the framework’s terminal-verdict for condensed-matter applicability. The paper’s value is in **specifying a falsifiable test that does not currently exist in the campaign’s repertoire** — the test is worth specifying whether or not the framework wins it, because at present no condensed-matter prediction of the framework is on the books at all.

— Trey